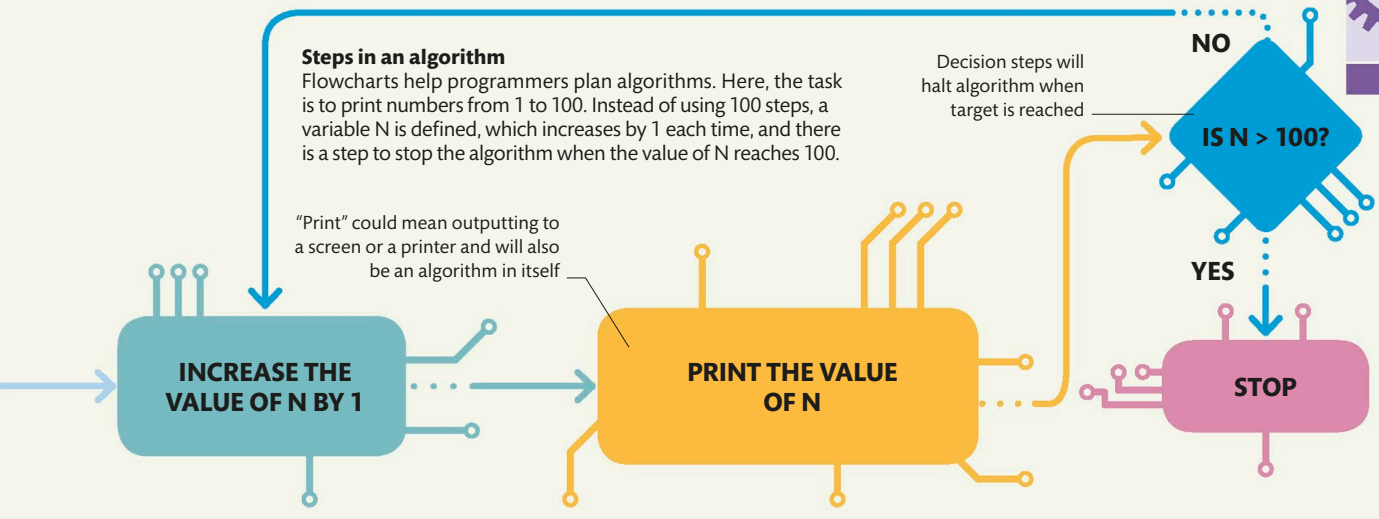
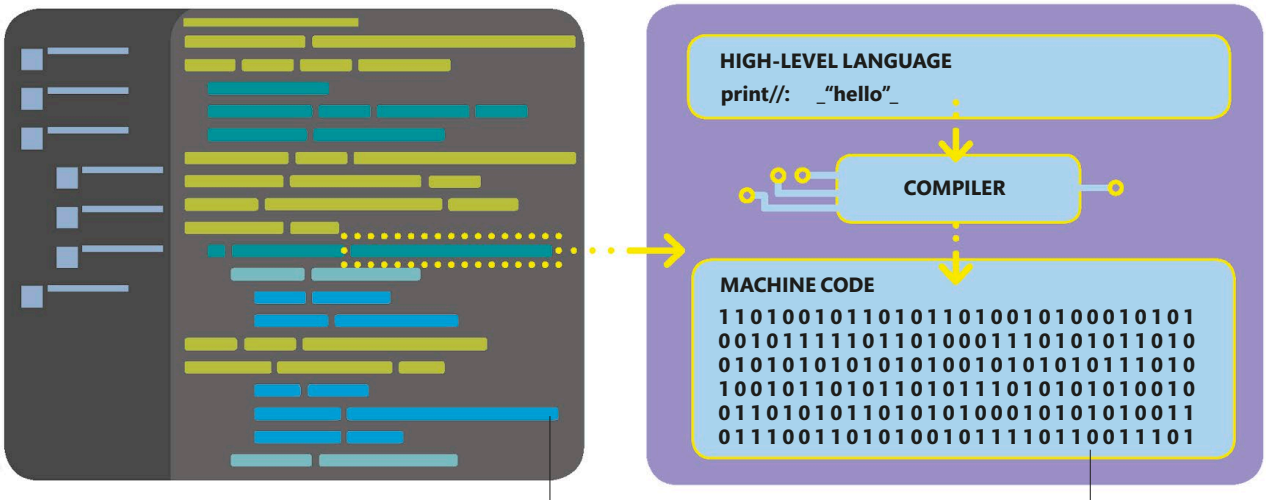
Steps in an algorithm
Flowcharts help programmers plan algorithms. Here, the task is to print numbers from 1 to 100. Instead of using 100 steps, a variable N is defined, which increases by 1 each time, and there is a step to stop the algorithm when the value of N reaches 100.





From high-level language to machine code
A compiler translates source code, written in high-level programming languages, into machine code. The result is an executable file consisting of binary numbers.

Application written, or coded, in high-level language

High-level language translated into machine code

Programs and code

Programs are written, or coded, in words and symbols that humans can read and write. These words and symbols are known as high-level languages—such as Java and C++. The full set of instructions that make up a program is known as the source code. A computer processor cannot understand high-level languages, only binary numbers. Source code is translated by a program called a compiler into a set of on-and-off electric currents in the memory and processor that represent binary numbers. This is called machine code.

THE COMPUTER IN **NASA'S SPACE SHUTTLE** USED **LESS CODE** THAN MOST OF TODAY'S **CELL PHONES**